MIL-E-1/348A
28 December 1967
SUPERSEDING
MIL-E-1/348
20 August 1953

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5Z4

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Rectifier, full-wave, high vacuum

Outline	---	8-6 (EIA) Ⓐ
Base	---	B5-25
Envelope	---	MT-8
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	4	6	8
Element	---	sh	h	2p	1p	h, k

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	epx	Epp	ib	Io	RL	CL	i(surge)	Alt
Unit:	Vac	v	Vac	ma	mAdc	Ohms	μf	a	ft
Maximum:	5.5	1,550	---	415	140	---	---	1.4	10,000
Minimum:	4.5	---	---	---	---	---	---	---	---
TEST CONDITIONS:	5.0	---	400	---	---	3,300	4	---	---

REQUIREMENT OR TEST:

GENERAL

3.2	Qualification - Required	E-50.3	Preheating
E-50.2	Holding period	5.	Preparation for delivery

Ⓐ denotes technical changes only.

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METHOD OR PARAGRAPH	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Quality conformance inspection, part 1							
1353	Operation of rectifiers	See notes 1 and 3	0.65	II	Io	133	---	mAdc
(A)1231	Emission (1)	E2b = 0; E1b = 50 Vdc	0.65	II	I1s	300	---	mAdc
(A)1231	Emission (2)	E1b = 0; E2b = 50 Vdc	0.65	II	I2s	300	---	mAdc
1201	Short and discontinuity detection		0.4	II	---	---	---	---
Appendix D, 20(a) 30	Visual and mechanical inspection criteria		---	---	---	---	---	---
Appendix D, 20(b)	Dimensions	Outline 8-6; dimensions A, B, C, D, L, and M	1.0	I	---	---	---	---
	Quality conformance inspection, part 2							
1111	Base pin solder depth		6.5	L6	---	---	---	---
1301	Heater current		6.5	L6	If	1.8	2.2	A
1101	Secureness of base and base insert		6.5	L6	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
4.7 and 1501	Life-test provisions (intermittent)	Group B (see note 2)	---	---	---	---	---	---
4.7.3	Life-test end point (intermittent) (500 hours)	Operation of rectifiers	---	---	Io	111	---	mAdc
	Periodic-check test (see 4.1.1.2)							
1031	Low-frequency vibration	No voltages	---	---	---	---	---	---

NOTES:

1. In a full-wave circuit adjust Zp/p such that a tube having Etd = 20 Vdc at 125 mAdc per anode gives Io = 140 mAdc. The tube may be preheated at any heater potential below 3.8 Vac.
2. In life-test operation, the values of RL and CL given in test conditions may be considered as approximate, and shall be adjusted initially to give Io = 140 mAdc with ib greater than 375 ma.
3. This test to be performed at the conclusion of the holding period.

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NOTES: - Continued

4. The tube described by this TSS is interchangeable with the tube procured to the previous issue. (A)

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL
Navy - EC
Air Force - 11, 85
DSA - ES

User activities:
Army - ME
Navy - AS, OS, MC, CG
Air Force - 19

Preparing activity:
Air Force - 11

(Project 5960-2247-57)

1968-301-516 3757

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